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The Design of the Recovery Electronics Module for the 2023 IREC Competition

A Report Prepared For:
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Dear Professor Owen,

This report, entitled “The Design of the Recovery Electronics Module for the 2023 IREC Competition”, was written to fulfil my 3A work term report requirements. This is my 2nd work report submission.

This report serves as the design report about how the recovery electronics module was designed for the recovery section of Waterloo Rocketry’s 2023 rocket Leviathan of the Sky (LotS). The purpose of this report is to fully document and explain all the factors leading to the final design of the avionics module. It contains documentation on the full design cycle, from conception and requirements to post flight thoughts and recommendations.

This report was written entirely by me and has not received any previous academic credit at this or any other institution. I would like to acknowledge the help of Jack Christensen, the former electronics subsystem lead. His knowledge and understanding of electrical systems were critical to optimizing the design and understanding the criteria on each individual electrical component. All the projects Waterloo Rocketry works on are fully funded by generous sponsors who donate financial or technical contributions. Without their generosity, this and many other projects would not be possible.

Best Regards

Stefan Arroyo-Cottier

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Summary

The rocket designed by Waterloo Rocketry for the 2023 IREC competition hosts many electronics spread out across the entire rocket. These control many aspects of the propulsion, payload, and recovery systems. The recovery systems in specific require a large portion of those electronics. As such, there is the need for a dedicated recovery electronics module.

Although an equivalent system has been designed for every rocket launched by the team, a lack of documentation means that there is the need to design and document a new avionics module. Pulling on experience from previous years, this assembly was designed for use in desert conditions with greater accessibility over other years. Furthermore, there are rigorous loading conditions that were analyzed to ensure the assembly would survive the shock of parachute deployment.

Many materials and configurations were analyzed, and the final design consisted of two FRP plates with rivet nuts installed to mount the electronics boards and other hardware. Aluminum L-brackets were used to fasten the assembly to its bulkhead. Furthermore, an antenna mount was designed, and 3D printed to minimize the RF interference of the assembly.

Ultimately, the design worked nominally, surviving two launches and a hard landing that inflicted upwards of 66G of deceleration on the rocket. This highlights the quality of the design. Feedback from the electrical subsystems noted improved assembly and improved wire routing.

1.0 Introduction

1.1 Waterloo Rocketry Team

The Waterloo Rocketry team is a student design team from the University of Waterloo. The team competes annually at the Launch Canada rocketry competition (but has competed in the Intercollegiate Rocketry Engineering Competition known as IREC for all previous launches). This design was for the 2023 rocket flown at IREC, the Leviathan of the Sky (LoTS), the LoTS is an entirely student designed rocket, flying with a hybrid engine to a target apogee of 30,000 ft [1]. It sports a scientific payload and an experimental reefed recovery system which has been in development for three years. The overall rocket cross section can be seen below in Figure 1.



Figure 1: Cross Section of the LoTS [1]

The team is subdivided into different subsystems, each with ownership over their respective domain/ section of the rocket. Most important for this report are the electrical and recovery subsystems. The electrical subsystem handles the design and manufacture of all electrical hardware and circuit boards before handing them off to the specific owners of each section for integration.

1.2 Recovery Subsystem

The recovery subsystem ensures a quick and safe descent from apogee. Due to the extreme height the rocket goes to, a two-stage parachute system is required to minimize drift and overall flight time. A small parachute known as the drogue chute deploys at apogee, stabilizing

the rocket and preparing for the deployment of the main chute, which deploys just before landing. The LotS flies a reefed parachute system, where the main parachute and drogue parachute are the same as seen below in Figure 2.

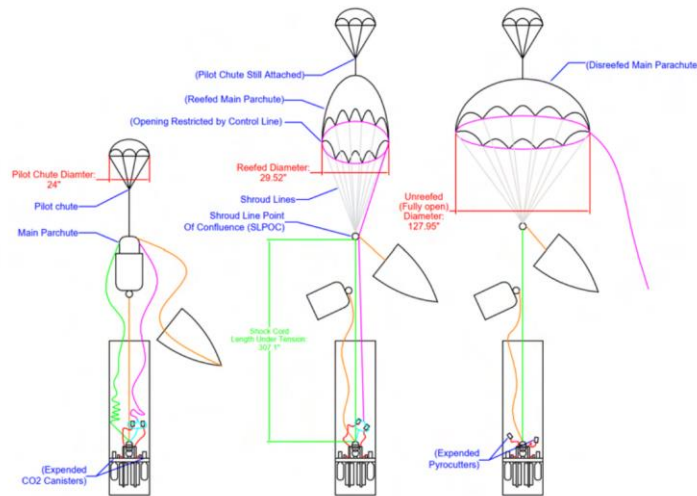


Figure 2: Recovery Concept of Events [1]

The drogue parachute is constrained by a reefing line and at the time of main parachute deployment, a set of pyrotechnic cutters activate, severing the cord and opening the parachute to its full size. As such, there is a need for a significant amount of electronics to be present within the recovery section. Beyond the payload and engine, the recovery section hosts a significant amount of electronics, from altimeters to control parachute deployment, to a logger board for data storage among other components. As such, a dedicated recovery electronics module is required to mount the electronics during flight. Due to the presence of other recovery specific hardware such as the main loaded eyebolt that ties to the parachute, a Co2 ejection system and various bulkhead passthroughs, the design of the recovery electronics mounting isn't as straight forward as it might seem. Each year, the electronics on board the rocket have increased in complexity [2] and as such, a new mounting solution is required.

1.3 Objective

The objective of this project is to design an accessible mounting solution for all required electronics hardware. Furthermore, it is critical to ensure all electronic components can function nominally as failure implies a descent without a parachute, which is exceedingly dangerous and is a loss of a significant amount of hardware. From a competition standpoint, the loss of recovery electronics would constitute a recovery failure and the loss of 30% of the flight points [3]. The project must resolve outstanding issues the electronics team has encountered concerning wire routing and board accessibility and swapping.

2.0 Problem

2.1 Problem Definition

Defining the problem follows from the objective, however finding the solution space is more complicated. There were a handful of driving constraints the design was based on: ease of manufacturing, ease of use in the field and ease of board replacement were the key features designed for. Ease of manufacturing is important as the components had to be able to be manufactured by a relatively unskilled student using the tools present in the Engineering Student Machine Shop. This meant that complex geometry must be avoided, although CNC machining is available it is a source of significant complexity. Furthermore, ease of field assembly is critical. Previous iterations of the sled had the circuit boards be mounted using standoffs, bolts, and nuts. M3 nuts and bolts have proven extremely difficult to assemble in windy desert conditions and they were to be avoided in the design. Finally, since many of the boards were student researched and designed, there was a significant chance that the boards would have to be swapped right before flight. As such, it was critical that the boards be easy to swap with minimal disassembly.

2.2 Past Work

The first step was to look at previous examples of what the team has done before. This would allow for the saving of time and effort by basing design choices on already defined analysis. Unfortunately, due to a lack of documentation from previous rocket iterations, little is known about other approaches to the recovery electronics sled within the team. A design report from 2016 [2] highlights a plate-based design approach with the final design pictured below in Figure

3.

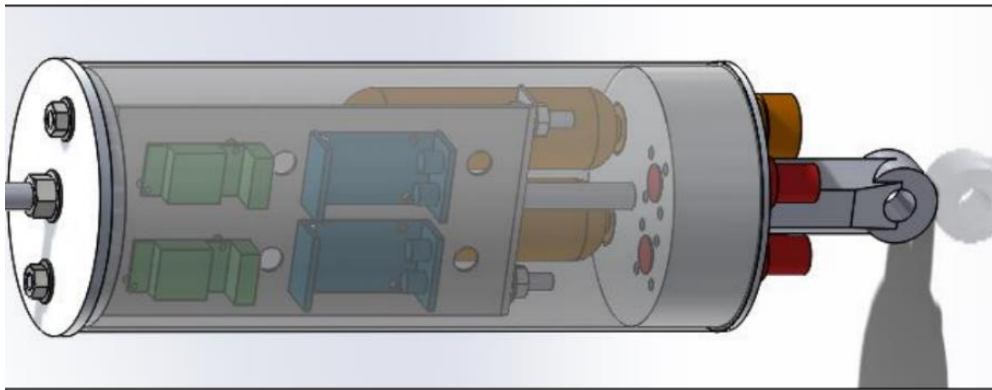


Figure 3: 2016 Avionics Bay [2]

Unfortunately, not much of the design is applicable as the diameter of the rocket has increased from 4" to the current 6". This means that there is little in the way of previous designs to work from and that much of the design had to be approached from scratch. Furthermore, the electronics payload has increased significantly in complexity. Although the bare minimum electronics required are two altimeters and two batteries as seen in the previous design, the addition of more ambitious projects from the electronics sub system means more real estate is required to house them.

2.3 Constraints and Criteria

The constraints and criteria were divided into an assembly and component level. The component level design criteria helped ensure that the design would allow each component to work optimally. The assembly level constraints and criteria ensured that the design would work nominally in its entirety. The criteria were defined as follows:

1. The assembly shall be easily removeable from the bulkhead on which it is mounted. Due to the architecture of the recovery section, this constraint allows for the electrical

assembly to be removed from the rest of the recovery hardware, this is critical for allowing boards to be swapped in the event of a failure before launch. Easy is defined by an untrained person being able to remove the bulkhead with simple hand tools in under three minutes.

2. Once removed from the rocket, all components shall be directly accessible and removable without disassembly of other components. This constraint helps define what easily removable components are, ensuring each component is individually accessible.
3. The removal of all components shall be completed using only hand tools. This constraint ensures that the removal of components is using simple tools (such as a hex key or screwdriver). This eliminates the need for specialized hardware when trying to swap components.
4. The assembly shall withstand accelerations of 40Gs laterally and 14Gs axially. From previous flights, we have collected accelerometer data that provides a rough estimate of the forces the rocket experiences at peak deceleration (upon parachute deployment). By taking these values and adding a safety factor of 1.5, loading criteria can be identified.
5. The assembly shall secure the following components [4]:
 - a. Remote arming board
 - b. Stratologger Altimeter
 - c. Raven 4 Altimeter
 - d. Two 9V battery holders
 - e. GPS board and antenna
 - f. radio board and breakout radio board

- g. live telemetry antenna
 - h. logger board
6. Must fit within a cylinder that has an inside diameter of 5.74 inches. The outside diameter of the rocket is 6 inches and the narrowest point through which the assembly must pass through during installation measures 5.8 inches. Having a small factor of safety helps keep components safe while installing them.

The constraints highlight a variety of quality-of-life changes for the design:

1. Reduce the length of the assembly as much as possible. This gives the attached antennas as much space as possible for a better connection. Furthermore, it helps reduce the overall length of the rocket, increasing stability.
2. Reduce the mass of the assembly as much as possible. The amount of weight on the rocket is directly proportional to the altitude achieved. As such, reasonable efforts must be made to keep every component as light as possible.

All the boards had similar criteria concerning enough space to mount the board and various connectors, these were determined from conversations with the electrical subsystem [4].

However, some boards had specific unique requirements that are outlined below in Table 1.

Table 1: Component Level Criteria

Component	Criteria
Remote Arming Board	• There must be a 2 cm margin around all edges of the board for connectors • Must be positioned close enough to the body tube for the magnetic switches to be activated from outside the rocket

Stratologger Altimeter	• Must be placed vertically along the long axis of the rocket • There must be one cm of clearance along each side of the length of the board
Raven 4 Altimeter	• Must be placed vertically along the long axis of the rocket • Must have one cm of clearance along the width of the board
9V Battery Holders	• Must be placed vertically along the long axis of the rocket with contacts facing down • Must have a 4cm clearance envelope to allow for battery mounting and securing • The contacts must have a 1cm clearance around the edge of the contacts to allow for wire bend radius
GPS Board and Antenna	• GPS antenna is to be placed at the edge of the assembly to avoid radio interference with other components
Radio Board and Breakout Board	• No particular requirements
Live Telemetry Antenna	• To be mounted at the bottom edge of the assembly, facing the motor with a minimum 5 inches of empty space below it for RF transparency
Logger Board	• No particular requirements
CO2 Canisters	• Require 1cm of clearance in each direction to allow for mounting and dismounting • Must be easily accessible with the rest of the assembly fully installed

3.0 Design

3.1 Design Approach

The recovery electronics section involves the work of both the recovery and electronics subsystems with the recovery subsystem taking the lead on the mechanical design and mounting solution for the boards. The electronics team drove the constraints and requirements of each board. Since the design for the sled was being worked on in parallel with the design of the actual boards that were to be mounted on the sled, a black box design approach was used. This design approach involved multiple meetings early in the design process where the mechanical constraints of each board were set, including orientation, screw hole position and electrical connector positions. This allows for each board to be simplified down to its physical volume and interactions with the rest of the system. This was then translated into a SolidWorks sketch where the position and orientation of each component could be manipulated to find the optimal configuration through an iterative approach.

3.2 Overall Architecture

The definition of the overall architecture of the electronics section took inspiration from the 2016 avionics bay. It was clear that the circuit boards are to be mounted on some sort of plate to maximize the space used. They cannot be mounted on the inside of the rocket airframe as that would violate the structural integrity of the composite tubes used. Initial design ideation for the plate positioning came up with two main designs. The main consideration was the orientation of boards for flight. The specifications for both altimeters indicated they must be placed along the axis of the rocket along with the battery holders. Furthermore, the

specifications for the remote arming board show that it must be placed close to the body tube so the magnetic arming switches can be appropriately activated. The first conceptual design was a set of cantilevered vertical plates that run along the length of the rocket that would accommodate electronics in the “inside” and “outside” sides of the plate. The second was multiple circular plates stacked on top of each other as seen in Figure 4. The benefit of the second design option was that it allowed for easy CO_2 canister access, with a small vertical section to accommodate the vertical components. Furthermore, this design involves the magnetic switches being taken off the remote arming board and positioned closer to the edge of the body tube, beside the altimeters. The main flaw with this design in the positioning of the battery holders, it does not take advantage of the direction of acceleration. This means it is liable to shake lose or potentially lose contact during flight.

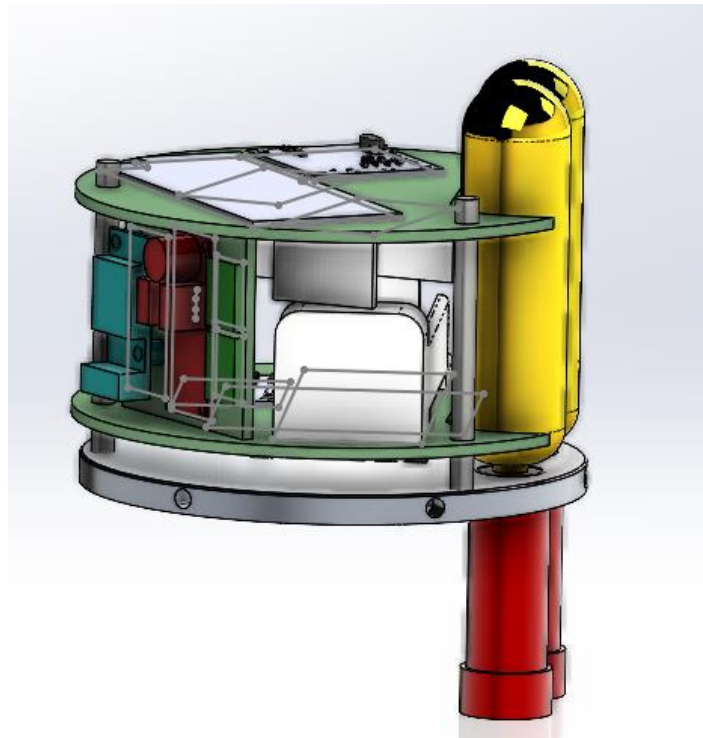


Figure 4: Proposed Plate Design

Furthermore, it is more difficult to access individual components, especially if the assembly is already mounted on the rocket. The cantilevered plates (or sleds as they are referred to within the team) were ultimately selected due to superior board accessibility and due to the ability for the entire sled to be easily detached from the mounting bulkhead. However, cantilevered plates had the concern of being structurally unstable potentially being vulnerable to lateral accelerations. They also have the additional concern that for a longer plate, the bending loads could increase to the point where the plate breaks or impacts the inside of the body tube. This in turn raised concerns of not only the plate breaking and the recovery section coming apart, but also of the vibrations causing the mounting bolts to come loose. As such, the following failure modes were identified:

1. Failure due to cantilever bending
2. Failure due to fasteners vibrating loose
3. Failure due to fasteners shearing out of place

3.3 Component Selection

With the overall architecture of the recovery electronics sled being already selected, the detail design and organization of the boards remained. Most of the detailed design was driven by the mounting solution that was selected for the boards. This in turn was driven by the selected material for the boards. The material selection of the boards came down to three main materials: plywood, aluminum and FRP (fiber reinforced plastic). These materials were selected due to their availability, relatively low cost, and ease of manufacturing. To identify the ideal material a selection matrix in Table 2 was used to weigh the different properties and

considerations of each material. Each property was weighted on a scale of 1-5 with one being the worst possible score and five being the best possible score. The total points of each material were then added and the material with the highest sum was selected.

Table 2: Material Selection Matrix

Consideration	Plywood	Aluminum Plate	Aluminum Sheet	Fiber Reinforced Plastic
Weight	3	2	3	4
Cost	5	3	4	3
Ease of Mounting	2	5	4	4
Manufacturability	3	5	5	5
Availability	5	5	5	4
Resistivity	5	1	1	5
Sum	23	21	22	25

Having selected FRP as the material of choice for the sled, there are multiple mounting methods that were trialed. The nut and bolt, tapping the FRP and a threaded clinch nut insert were selected for testing. Due to previous experience with trying to mount boards using nuts and bolts, it was decided that this mounting method was to be avoided. M3 hardware is too small to be changed reliably in field conditions. Furthermore, the number of fasteners required is 30+ due to each circuit board having approximately 4 mounting points. Tapping the FRP was attempted, and the results indicated that tapped FRP plate could withstand upwards of 30lbs of force before the internal thread stripped. However, attempting to replace the screw to simulate a board being changed showed that the hole would strip and lose its threads after just 2-3 attempts, meaning

it was unsuitable. The final mounting solution was to use a clinch nut on the FRP plates. This was ultimately selected as it provided many benefits, creating a permanent threaded hole in the FRP. Furthermore, this made assembly extremely easy as a screw just had to be driven into the plate. This also had the advantage of being easy to manufacture. The ESMS has a waterjet machine that allows for the plates to be quickly and easily cut to size. The required holes for the clinch nuts were also cut out on the waterjet, meaning sled manufacturing had only two steps, water jetting and clinch nut installation. Due to the ease of the waterjet cutouts, the design was improved to include multiple cutouts for cable routing. These would allow for small zip ties to mount the wires directly into the board. This is an improvement over the previous design, where 3D printed wire mounts were epoxied directly onto the board.

3.4 Failure Modes Analysis

1. Failure due to cantilever bending

The main issue due to cantilever bending occurs at the moment of parachute deployment. The parachute deploys while the rocket is at the peak of its parabola meaning it has significant lateral velocity. Upon parachute deployment the rocket undergoes a significant acceleration to return to the vertical orientation under the parachute. These accelerations have been recorded to up to 26G with a design acceleration of 40Gs. By checking the SDS sheet of the FRP plates which were ordered off McMaster-Carr [5] it is evident the manufacturer of the product is BEDFORD reinforced plastics. A material datasheet by the manufacturer places the modulus of elasticity at 19.3 GPa [6]. This means that the cantilever bending of a singular plate can be

found by the application of a distributed load to represent the mass of the assembly. The formula for maximum deflection in a cantilevered by with a distributed load is as follows:

$$Deflection = \frac{q * L^4}{8 * E * I}$$

Where q is the distributed load, L is the length of the plate, E is the elastic modulus, and I is the moment of inertia. The distributed load was found by taking the mass of the assembly (measured to 0.657kg) and multiplying it by the acceleration of gravity and the gravity multiplier). These values are as follows:

$$q = \frac{0.657kg * 9.81 * 40}{L}$$

$$L = 165 * 10^{-3}m$$

$$E = 19.3GPa$$

$$I = \frac{125 * 10^{-3}m * 3.5 * 10^{-3}m}{12} = 4.45 * 10^{-10}m^4$$

As such, the estimated deflection is 16.8mm which indicates the possibility of failure at parachute deployment, either by hitting the inside wall or other sled.

To address these concerns a design change was modified. A spacer was installed between both plates to increase their stiffness and resistance to bending. This spacer joined both boards and made them the equivalent of an I-beam. An I-beam calculator [7] was used to calculate the new moment of inertia as in Figure 5.

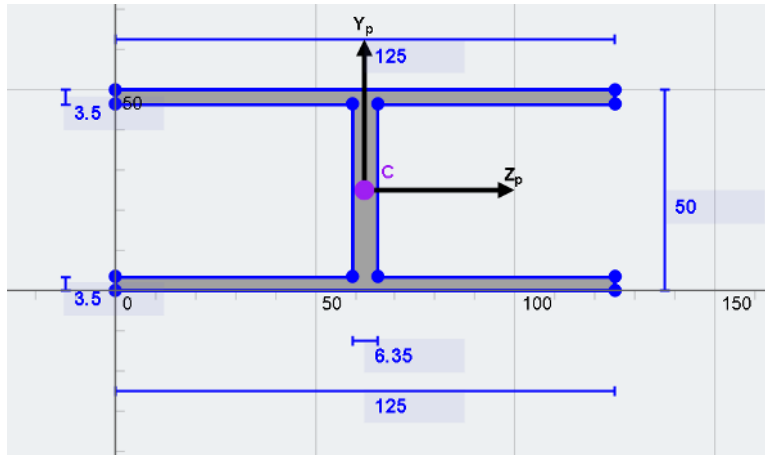


Figure 5: Stiffener Moment of Inertia [7]

This change increased the moment of inertia to $5.16 * 10^{-7} m^4$ which is three orders of magnitude larger than before. This in turn means a three order of magnitude reduction in deflection which is acceptable.

2. Failure due to fasteners vibrating loose

To resolve the issue of bolts vibration loose, the application of Loc-Tite was added to the assembly instructions, ensuring each bolt used had an adequate application of Loc-Tite. This is the standard operating procedure within the team and all flight hardware is Loc-Tited before flight and during final installation. Since there has never been a vibration induced bolt failure within the author's knowledge of the team, this solution was considered adequate.

3. Failure due to fasteners shearing out

Given the loading scenarios from previous flights, it is known that the largest deceleration will occur at the moment of initial parachute deployment. The force exerted by the inertia of each individual board will act on the component level fasteners which will in turn transmit the force onto the fasteners holding the assembly in place. As such, if the fasteners holding the assembly

in place are strong enough, then the fasteners holding the components in place shall suffice (all other things equal). To analyze this completely, two cases must be investigated: FRP board failure and bracket failure.

To convert the acceleration values from previous flights, the weight of the system is required. All of the components and fasteners were placed on a scale and weighed with a total weight of 1.448 lbs [0.657 kg]. The L-bracket attachment design was then put through an FEA study as in Figure 6. The loading condition was found by taking the mass of the system and multiplying it by the defined maximum acceleration of 40Gs.

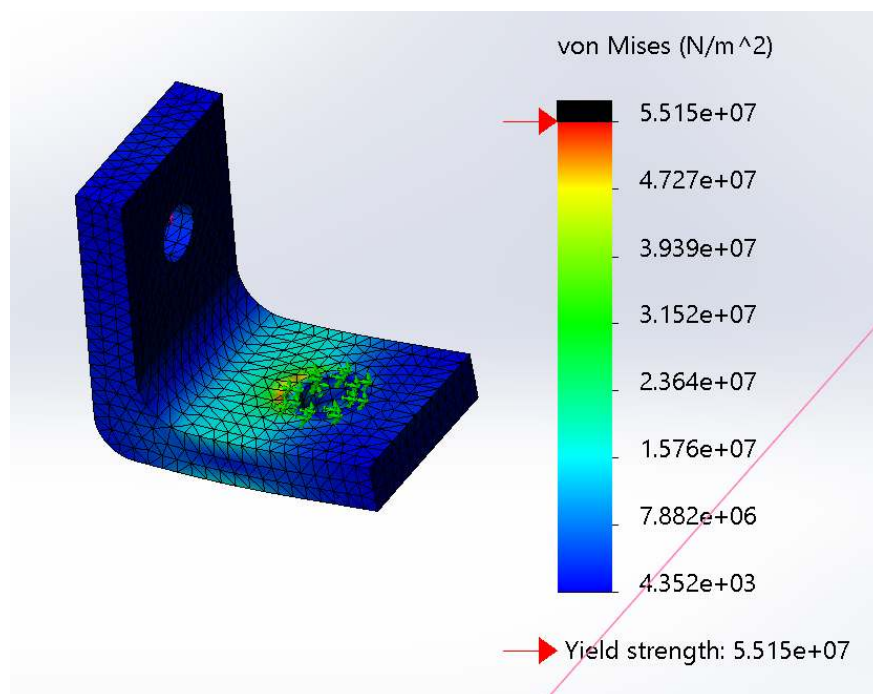


Figure 6: FEA of L-Brackets

Although there is some stress concentration where the edge of the bolt head meets the inside edge of the bracket, the bracket is sufficiently designed for this case.

The paired failure mode would be the rivet nut tearing out of the FRP plate. The assembly is pictured below in Figure 7 for a greater understanding of the interaction between the plate and the L-bracket.

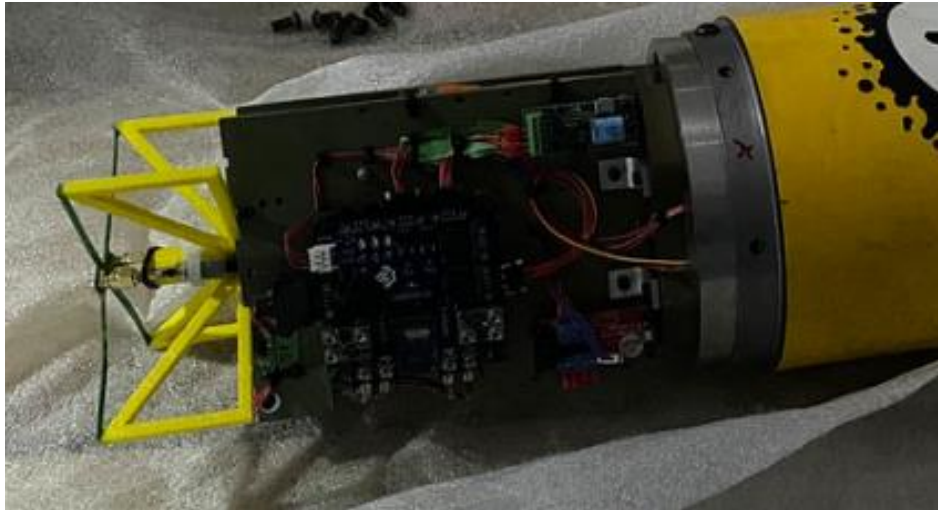


Figure 7: L-Brackets Installed in Assembly [8]

The tear out would be a shear failure from two shear planes, one on either side of the rivet nut. Since the FRP would behave more like a ceramic than a metal that can shear [9], the application of shear and thereby failure would be through crack propagation. As such the composites rule of thumb to have a hole two times the diameter away from the edge was used.

3.4 Manufacturing

Due to the selection of components and overall design, the manufacturing of the sled was straight forward but brought up some potential design issues. Having the rivet nuts meant they were a bit of a nub sticking out the end, where the thread was located. This interfered with the positioning of components that were flush with the FRP plate, specifically the battery holders. This meant their position had to be adjusted to avoid interference. Since all the electrical

boards were mounted on standoff, this did not affect them. As seen below in Figure 8 and Figure 9, both sides of a rivet nutted plate are visible. The mounting side is flush and the back side projects outwards approximately 3mm.



Figure 8: Back Side of Plate 1



Figure 9: Front Side of Plate 1

3.5 Final Design

The final design of the assembly includes some extra hardware not fully explained in the preceding portion of the report, namely the mount for the antenna. The author of this report (who is designer of the recovery electronics sled) had been operating under the assumption that the antenna hardware used would be two long antenna that would mount directly onto the live telemetry breakout board. However, the electrical team developed a new flat antenna to use. As such it necessitated the development of a mount for the cross shaped antenna. The first iteration was a simple extrusion of the footprint of the antenna downwards approximately 3mm in thickness. Subsequent feedback from the electrical team indicated there were high levels of interference with the metal present within the sled. The second iteration had lower but still unacceptable levels of interference with the 3d printed material of the mount. As such the final design removed as much material as possible. Due to the anisotropic properties of the 3D print, various attempts at FEA and hand calculations were unsuccessful. Since the antenna weighed only a couple grams, the antenna was glued, and zip tied in place making sure the connectors were secure. The final design can be seen below in Figure 10.

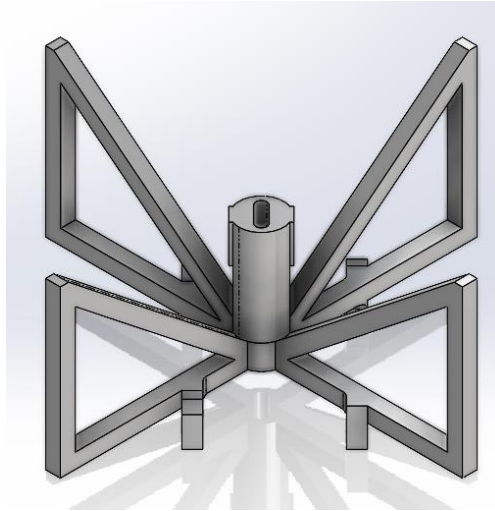


Figure 10: Final Iteration of Antenna Mount

The final design can be seen below in Figure 11. It is mounted on a 3D printed bulkhead for test and fit purposes. Note the integration of the antenna mount and harnessing of the wires.

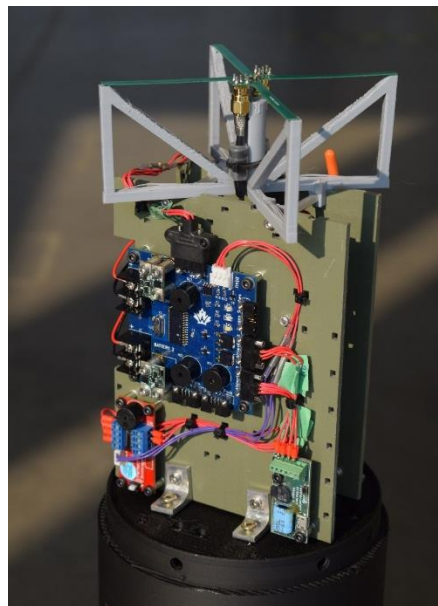


Figure 11: Finished Rec Elec Sled [8]

3.6 Post Competition Analysis

This recovery electronics sled flew two times, once at the IREC competition in New Mexico in 2023 and again at the Launch Canada competition in Ontario. Out of both competitions, the recovery electronics sled performed nominally and was found after recovery to be immediately reusable. The consensus from the electronics team was overall satisfaction with no comments for improvements from a quality-of-life standpoint. A recovery failure meant that the rocket impacted the ground at high velocity exerting over 66Gs in the process. The survival of the assembly speaks to the robustness of the design.

4.0 Conclusions

4.1 Conclusions

A new system for the mounting and assembly of electronic hardware was ideated and developed. Using team knowledge, FEA and hand calculations, critical failure modes were identified and analyzed. Furthermore, significant quality of life improvements was made on behalf of the electronics team, making the mounting of boards easier and adding built-in wire routing channels for a cleaner design. Accessibility was maintained at all levels of the assembly from individual components being easy to remove to the entire assembly easily detaching from the bulkhead on which it is mounted.

The overall design can easily be extended and modified to accommodate different boards on future rockets. The system's robustness and failure tolerance were extensively (unintentionally) tested through two crash landings which caused decelerations of over 66Gs. With no failures so far, this can serve as the base on which new recovery electronics sleds are developed.

4.2 Next Steps / Recommendations

The largest gap in the analysis comes from the interactions between the rivet nuts and the fiber reinforced plastic board. Further analysis would benefit from attempting to better characterize the material under the specific loading conditions. Two options are immediately available, a typical tensile test or a more representative tear out test with a rivet nut installed. This allows for more confidence and the ability to move boards closer to the edge of the FRP without fear of failure. Finally, the assembly would benefit from being placed on a shaker tuned to the vibrations

of the engine. This would ensure that vibrations cannot damage the sled in any way. This can also be approached from a mechanical solution removing any degree of motion still available to the sled.

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